

Graphical abstract legend and explanation of the analysis scripts

This document explains the graphical abstract and the accompanying public analysis scripts for the current myofibre density (*MFD*)-based analysis of steroid-naïve Duchenne muscular dystrophy (DMD) muscle biopsy specimens. It follows the visual and computational workflow shown in the figure and is intended as a guide to specimen selection, digital restoration, myofibre enumeration, statistical modelling and biological interpretation.

The upper part of the graphical abstract details the origin of the samples. The current analysis uses archival quadriceps femoris biopsy slides from steroid-naïve boys with DMD diagnosed at the former Yakumo National Hospital between 1973 and 1984. Because clinical documentation was incomplete, preliminary case identification partly relied on historical recollections, followed by review of available records and pathological evaluation. In total, 46 biopsies were screened by three pathologists; 8 were excluded because of severe fading or insufficient tissue quantity, leaving 38 analysable patients aged 1–16 years. Haematoxylin-eosin (H&E), Gömöri trichrome (G-T) and Masson trichrome (M-T) slides formed the primary material, whereas actomyosin adenosine triphosphatase (ATPase)-stained slides were reserved for sensitivity analyses.

The middle part of the graphical abstract explains why digital restoration was essential rather than cosmetic. Image acquisition used a Nikon ECLIPSE Ci microscope with LED illumination, a Nikon PLAN APO λ D 40 $\times$ objective, a Nikon DS-Fi3 camera and Nikon NIS-D software. Fields of interest were selected by consensus among three pathologists. High-resolution images were captured at 400 $\times$ magnification, with $2,880 \times 2,048$ pixels per frame and a spatial resolution of 0.11 $\mu\text{m}/\text{pixel}$. Sixteen tiled frames were then assembled into composite images of $11,520 \times 8,192$ pixels, corresponding to approximately 1.14 mm². Look-up table (LUT)-based restoration in Nikon NIS-D, colour adjustment, image tiling, monochrome conversion and Focus Stacking in Adobe Photoshop were applied to recover analysable histological information from faded slides and uneven focal planes.

In the scripts, the restored histological images are translated into case-level quantitative variables. Individual myofibres were traced in Adobe Photoshop 2024 using the Magnetic Lasso Tool with contrast 40 and frequency 100, and the resulting measurements were exported to Microsoft Excel and then saved as CSV files for statistical analysis in R version 4.3.2. *MFD* was defined as the number of myofibres per mm². Truncated fibres at the field edge were excluded regardless of central nucleation, hypercontraction or other pathological features. *MFD* was generated during complete enumeration of individual myofibres, while other parameters included mean myofibre size, standard deviation of myofibre size, coefficient of variation, myofibre area, connective/fibrotic tissue area, fatty degeneration area, necrotic fibre area, regenerative fibre area, opaque fibres relative to *MFD* and internally nucleated fibres relative to *MFD*.

The analytical intention of the first script block is quality control across staining methods.

Because H&E and G-T slides were combined in the primary dataset, their comparability was tested before subsequent modelling. In 28 paired H&E and G-T samples, myofibre area showed no meaningful difference by Wilcoxon signed-rank testing ($p \approx 0.9$), with negligible effect size (Cliff's $\delta = 0.015$ [95% confidence interval (CI): $-0.28, 0.31$]) and extremely strong concordance (Pearson's $r = 0.970, p < 0.001$). Bayesian paired analysis placed approximately 82% of the posterior distribution within the region of practical equivalence, with $BF_{10} = 0.101$. Mixed-effects modelling attributed only 0.03% specimen-level variance reduction to staining method. These script-level checks supported practical integration of H&E and G-T measurements, although broader validation involving M-T and frozen-versus-formalin-fixed comparisons remained limited by slide availability.

The lower-left part of the graphical abstract shows the main transition from morphometry to statistical modelling. In the scripts, correlation screening first identified *MFD* as the strongest inverse correlate of age at biopsy (Spearman's $\rho = -0.85$). Other significant correlations included mean myofibre size ($\rho = 0.61$), standard deviation of myofibre size ($\rho = 0.74$), coefficient of variation ($\rho = 0.66$), myofibre area ($\rho = -0.60$), fatty degeneration area ($\rho = 0.69$), connective/fibrotic tissue area ($\rho = 0.48$) and opaque fibres relative to *MFD* ($\rho = 0.34$). After log transformation, the association between age at biopsy and *MFD* remained strong and more linear (Pearson's $r = -0.86$), supporting *MFD*-centred regression modelling.

In the scripts, multiple regression was used as an exploratory bridge between raw morphometric measurements and age-linked tissue dynamics. A model including *MFD* and the *MFD* $\times$ standard deviation interaction explained approximately 80% of age-related variance, with adjusted $R^2 = 0.79$, Akaike information criterion (AIC) = 53.84 and leave-one-out cross-validation root mean squared error (RMSE) = 0.49. The standardised coefficient for *MFD* was -0.87 [CI: $-1.03, -0.72$], and the *MFD* $\times$ standard deviation interaction was 0.21 [CI: $0.080, 0.35$]. When the analysis was restricted to patients younger than 7 years, model fit improved markedly (adjusted $R^2 = 0.75$, AIC = 29.12), while predictive accuracy remained comparable (RMSE = 0.47). This script block therefore showed that early-stage histopathology was more coherent before approximately 7 years, whereas later-stage pathology became more heterogeneous.

To explain the *MFD* distribution of the graphical abstract, *MFD* showed a stepwise decrease across early age bands. Median *MFD* was 1109.62 fibres/mm² at 1–4 years, 765.96 at 4–6 years, 383.21 at 6–7 years, 451.66 at 7–9 years and 342.28 at 9–11 years. The contrast between 1–6 and 6–7 years was large (Glass's $\Delta = -1.75$ [CI: $-2.85, -1.25$]; mean difference -495.60 fibres/mm² [CI: $-682.45, -308.75$]), as was the contrast between 1–6 and 7–11 years (Glass's $\Delta = -1.81$ [CI: $-2.91, -1.35$]; mean difference -512.81 fibres/mm² [CI: $-689.04, -336.58$]). A representative comparison explains why the figure emphasises a counting-based index: a 4-year-4-month case and a 6-year-2-month case had almost identical myofibre area values (65.69% and 65.77%) and similar connective/fibrotic tissue area values (32.06% and 31.55%), whereas *MFD* declined from 765.96 to

454.67 fibres/mm². Thus, *MFD* detected early tissue depletion that area-based indices could visually and quantitatively miss.

The analytical intention of the logistic-regression script block is to examine whether an *MFD* range could distinguish patients below and above age 6. Significant *MFD* thresholds appeared between 546 and 572 fibres/mm². The most robust threshold range was 560–563 fibres/mm², with AIC = 19.19, classification accuracy = 94.29% and odds ratio = 0.0040 [CI: 0.00010, 0.043]. However, age 6 in this analysis remained a reference point selected for modelling rather than a statistically estimated breakpoint. For that reason, segmented regression was then used to treat the critical age itself as the estimand.

The segmented-regression part of the graphical abstract summarises the core result. In patients younger than 11 years ($n = 35$), segmented regression identified an *MFD* breakpoint at 6.25 years [CI: 5.08, 7.42]. Before this breakpoint, *MFD* declined steeply at -170.68 *MFD*/year [CI: -213.41 , -127.95]. After the breakpoint, the slope became much flatter at -28.31 *MFD*/year [CI: -76.47 , 19.85]. The segmented model had adjusted $R^2 = 0.80$ and improved significantly over a simple linear model ($F(2, 31) = 10.37$, $p < 0.001$). Leave-one-out cross-validation yielded a mean breakpoint of 6.26 years, and changing the initial breakpoint seed from 4 to 8 years still converged on 6.25 years. Replacement of primary *MFD* values with measurements from 14 ATPase-stained slides preserved the same pattern, with $\rho = -0.87$ and a similar breakpoint of 6.42 years [CI: 5.28, 7.56].

In the scripts, Bayesian and Monte Carlo analyses were used to test whether the segmented pattern remained stable under different assumptions. With the breakpoint fixed at 6.25 years, Bayesian segmented regression estimated a pre-breakpoint slope of -170.44 *MFD*/year [95% credible interval (CrI): -207.47 , -133.58] and a post-breakpoint slope of -28.42 *MFD*/year [CrI: -65.22 , 8.77]. Using a measurement-based region of practical equivalence of ± 62.00 *MFD*/year, 0% of pre-breakpoint posterior estimates and 96.26% of post-breakpoint estimates fell within that range. Bayesian model comparison favoured segmented regression over single-line regression, with Bayesian information criterion values of approximately 455.82 versus 470.20 and an approximate BF_{10} of 1324. A second Bayesian model with weakly informative priors estimated a breakpoint of 6.37 years [CrI: 5.24, 7.66], again supporting rapid early decline followed by practical flattening.

Monte Carlo simulation translated this statistical breakpoint into an uncertainty scale. With 2×10^6 replicates, mean misclassification around the 6.25-year boundary was 38.92% when the true age gap was only 0.25 years, 12.99% at 1.00 year, 7.95% at 1.25 years and 1.21% at 2.00 years. Sample-size simulations showed that the current $n = 35$ dataset had 72.43%, 79.47% and 84.54% probabilities of detecting the breakpoint within ± 1.00 , ± 1.25 and ± 1.50 years, respectively. An inverted logistic model further defined *MFD* zones: ≥ 595.51 fibres/mm² corresponded to an approximately 80% probability of being younger than 6.25 years, 426.36–595.51 fibres/mm² defined a grey zone and ≤ 426.36 fibres/mm² corresponded to an approximately 80% probability of being 6.25 years or older.

The present findings suggest that early DMD pathology may include substantial myofibre loss before the more visually obvious expansion of connective tissue, fatty replacement and architectural disruption becomes dominant. *MFD* is not merely another morphometric variable; it is a complete fibre-counting readout that captured a rapid early decline partly masked by myofibre hypertrophy and by area-based measurements. These findings align histopathological change with the earlier clinical symptom period around 3–5 years and with functional transitions often observed around 6–7 years, while also challenging the assumption that early muscle architecture remains relatively stable until that interval.

The interpretation remains intentionally cautious. The *MFD*-based zones and the proposed therapeutic window are hypothesis-generating histopathological categories, not validated clinical treatment rules. The current analysis is retrospective, single-centre, based on archival material and limited by incomplete clinical records, heterogeneous staining methods and modest sample size. Whether low post-threshold *MFD* is irreversible, modifiable by corticosteroids or newer therapies, or usable as a clinical-trial endpoint requires longitudinal and multicentre validation. At the same time, the scripts demonstrate that digitally restored archival slides can generate reproducible quantitative pathology data, and that *MFD* may serve as a practical pathology-based reference point for future clinicopathological, imaging, radiomics and molecular validation studies.

Note:

1. The peer-reviewed *MFD*-based paper reports the central published result of this analysis: Myofibre Density Reveals a Critical Threshold around Age 6 in Steroid-Naïve Duchenne Muscular Dystrophy: A Retrospective Observational Study. PMCID: PMC12997517.
2. Reproducibility materials: datasets and R scripts are available in this GitHub branch.
3. Official HUSCAP record: link to be added after public release.